2-4

68 Prove that if A is an $m \times n$ matrix and P is an invertible $m \times m$ matrix, then $\operatorname{rank} PA = \operatorname{rank} A$.

Hint: $P^{-1}PA = A$

① $\operatorname{rank} PA \le \operatorname{rank} A$

② $\operatorname{rank} A = \operatorname{rank} \underbrace{P^{-1}PA}_{=A} \le \operatorname{rank} PA$

$\operatorname{rank} A \le \operatorname{rank} PA \le \operatorname{rank} A$

c.f. 2-3 62 Prove that if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then $\operatorname{rank} AB \le \operatorname{rank} B$.

69 Let B be an $n \times p$ matrix. Prove the following:

rank RB ≤ rank B

(a) For any $m \times n$ matrix R in reduced row echelon form, $\boxed{rank\ RB \leq rank\ R}$.

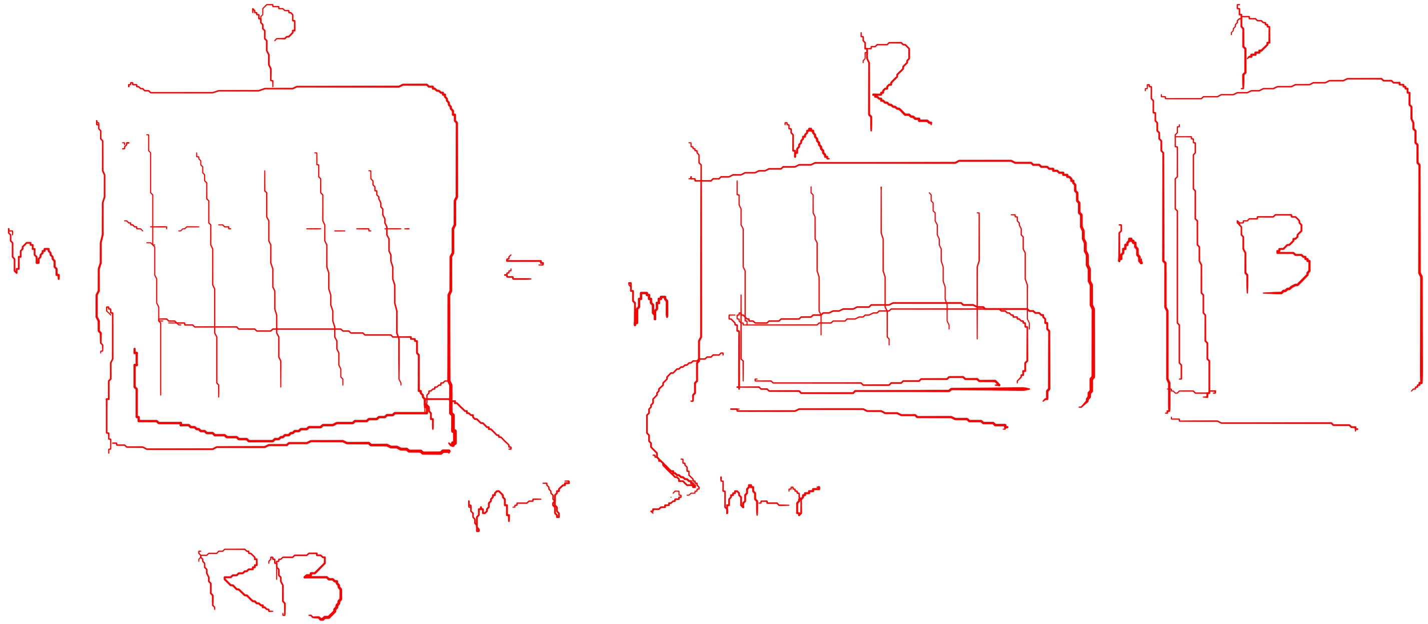
?

Hint: Consider the characteristics of RREF (zero rows).

RB's zero row ≥ R's zero row

c.f. 2-3 62 Prove that if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then $rank\ AB \leq rank\ B$.

c.f. 68 Prove that if A is an $m \times n$ matrix and P is an invertible $m \times m$ matrix, then $rank\ PA = rank\ A$.


p
m
=
m-r
RB
R
n
m
m-r
n
p
B

69 Let B be an $n \times p$ matrix. Prove the following:

(a) For any $m \times n$ matrix R in reduced row echelon form, $rank\ RB \leq rank\ R$.

(b) If A is any $m \times n$ matrix, then $rank\ AB \leq rank\ A$. $= rank R$ $A = PR$

inv. A's RREF

rank AB = rank PRB

= rank RB $\leq$ rank R = rank A

c.f. 2-3 62 Prove that if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then $rank\ AB \leq rank\ B$.

c.f. 68 Prove that if A is an $m \times n$ matrix and P is an invertible $m \times m$ matrix, then $rank\ PA = rank\ A$.

70 Prove that if A is an $m \times n$ matrix and Q is an invertible $n \times n$ matrix, then $rank\ AQ = rank\ A$.

$AQQ^{-1} = A$

① $rank\ AQ \leq rank\ A$

② $rank\ A = rank\ \underbrace{AQQ^{-1}}_{=A} \leq rank\ AQ$

$rank\ A \leq rank\ AQ \leq rank\ A$

c.f. 69 (b) If A is any $m \times n$ matrix, then $rank\ AB \leq rank\ A$.

2-3 62 Prove that if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then $rank\ AB \leq rank\ B$.

If A is inv, $rank AB = rank B$

68 Prove that if A is an $m \times n$ matrix and P is an invertible $m \times m$ matrix, then $rank\ PA = rank\ A$.

If B is inv, $rank AB = rank A$

69 (b) If A is any $m \times n$ matrix, then $rank\ AB \leq rank\ A$.

70 Prove that if A is an $m \times n$ matrix and Q is an invertible $n \times n$ matrix, then $rank\ AQ = rank\ A$.

$rank AB \leq \underline{min}(rank A,\ rank B)$

71 Prove that for any matrix A, $rank\ A^T = rank\ A$.

rank $R = r$

rank $A = r$

Hint: $A = PR$, P is invertible, R is RREF

$rank\ A^T = rank\ (PR)^T$

$= rank\ R^T P^T = rank\ R^T = rank\ R = rank\ A$

P^T inv.

c.f. 2-3 87. Let R be an $m \times n$ matrix in reduced row echelon form with $rank\ R = r$. Prove the following:

(a) The reduced row echelon form of R^T is the $n \times m$ matrix $[\boldsymbol{e}_1 \quad \cdots \quad \boldsymbol{e}_r \quad \boldsymbol{0} \quad \cdots \quad \boldsymbol{0}]$, where $\boldsymbol{e}_j$ is the jth standard vector of R^n for $1 \leq j \leq r$.

$rank\ (RREF(R^T)) = r$

(b) $rank\ R^T = rank\ R = r$

$$\operatorname{rank} A = n = \operatorname{rank} B = n$$

$$\underset{n\times n}{A} \text{ inv.} \qquad \underset{n\times n}{B} \text{ inv.}$$

$$\operatorname{rank} \underbrace{\overset{n\times n}{AB}}_{=n} = \underbrace{\operatorname{rank} A}_{=n} = \underbrace{\operatorname{rank} B}_{=n}$$

71 Prove that for any matrix A, $rank\ A^T = rank\ A$.

Hint: $A = PR$, P is invertible, R is RREF

c.f. 2-3 87. Let R be an $m \times n$ matrix in reduced row echelon form with $rank\ R = r$. Prove the following:

(a) The reduced row echelon form of R^T is the $n \times m$ matrix $[\boldsymbol{e_1} \quad \cdots \quad \boldsymbol{e_r} \quad \mathbf{0} \quad \cdots \quad \mathbf{0}]$, where $\boldsymbol{e_j}$ is the jth standard vector of R^n for $1 \leq j \leq r$.

(b) $rank\ R^T = rank\ R$

72 Use the Invertible Matrix Theorem to prove that for any subset S of n vectors in R^n, the set S is linearly independent if and only if S is a generating set for R^n.

$S = \{a_1, a_2 \cdots a_n\}$

$A = [a_1 \cdots a_n]$

$A: n \times n$

- Let A be an n x n matrix. A is invertible if and only if
  - The columns of A span R^n
  - For every b in R^n, the system Ax=b is consistent
  - The rank of A is n
  - The columns of A are linear independent
  - The only solution to Ax=0 is the zero vector
  - The nullity of A is zero
  - The reduced row echelon form of A is I_n
  - A is a product of elementary matrices
  - There exists an n x n matrix B such that BA = I_n
  - There exists an n x n matrix C such that AC = I_n

onto

One-to-one

= square matrix

73 Let R and S be matrices with the same number of rows, and suppose that the matrix $[R \quad S]$ is in reduced row echelon form. Prove that R is in reduced row echelon form.

82 Use matrix transposes to modify the algorithm for computing $A^{-1}B$ to devise an algorithm for computing AB^{-1}.

$$\left[\begin{array}{c|c} A & I \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{c|c} I & A^{-1} \end{array}\right]$$

$$\left[\begin{array}{c|c} A & B \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{c|c} I & A^{-1}B \end{array}\right]$$

$$(AB^{-1})^T = (B^{-1})^T A^T = (B^T)^{-1} A^T$$

$$\left[\begin{array}{c|c} B^T & A^T \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{c|c} I & (B^T)^{-1} A^T \end{array}\right]$$

$$= (AB^{-1})^T$$

T

83 Let A be an $m \times n$ matrix with reduced row echelon form R. $a_j = Ae_j$

(a) Prove that if $rank\ A = m$, then there is a unique $m \times m$ matrix P such that $PA = R$.

Hint: There exists an invertible $m \times m$ matrix P such that $PA = R$, so $A = P^{-1}R$

$m \times$

$$PA = R \Rightarrow {}^{m}\overset{n}{A} = P^{-1}R \qquad \text{exist } e_1 \ e_2 \ \cdots \ e_m = r_{i_1} \ = r_{i_2} \ \cdots \ = r_{i_m}$$

$$[a_1 \ a_2 \ \cdots \ a_n] = P^{-1}[r_1 \ r_2 \ \cdots \ r_n]$$

$$p^{-1}r_1 \quad p^{-1}r_2 \quad p^{-1}r_n$$

$$P^{-1} = [a_{i_1} \ a_{i_2} \ \cdots \ a_{i_m}] \rightarrow P$$

83 Let A be an $m \times n$ matrix with reduced row echelon form R.

rank $A = m$ unique

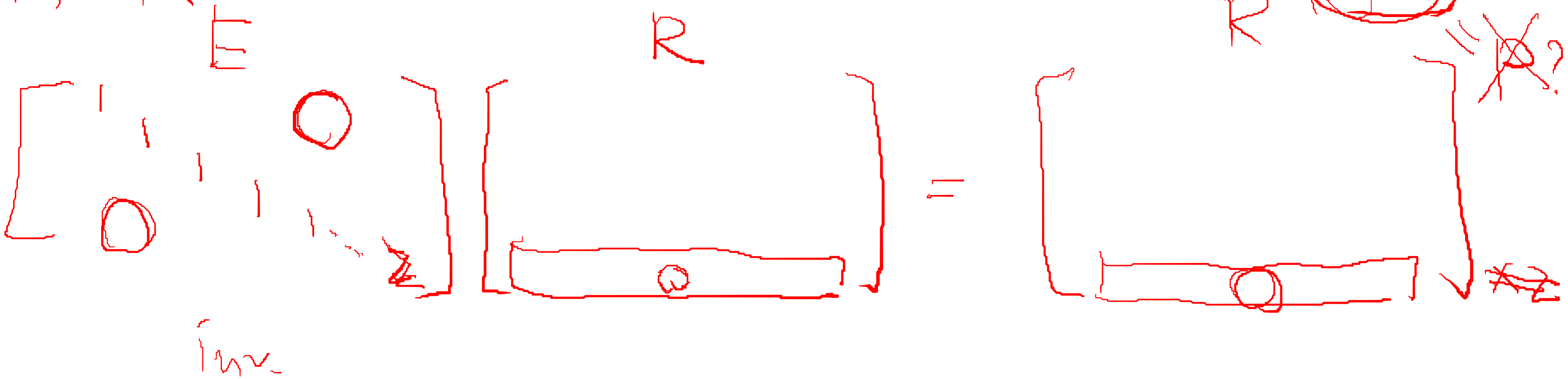
(b) Prove that if $\boxed{rank\ A < m}$, then there is more than one invertible matrix P such that $PA = R$.

Hint: Because $rank\ A < m$, R has zero row. There is an elementary $m \times m$ matrix E, distinct from I_m, such that $ER = R$.

$PA = R$

$PA = R$

$\Rightarrow EPA = ER = R$



Let A and B be $n \times n$ matrices. We say that A is **similar** to B if $B = P^{-1}AP$ for some invertible matrix P.

B is similar to A

87 Suppose that A and B be $n \times n$ matrices such that A is similar to B. Prove that A^T is similar to B^T.

$$B^T = (P^{-1}AP)^T$$

$$= P^T A^T (P^{-1})^T$$

$$= (P^T) (A^T) (P^T)^{-1}$$

$$B = P^{-1} A P \quad \text{inv.}$$

$$A = P B P^{-1} \quad \text{inv.}$$

Let A and B be $n \times n$ matrices. We say that A is **similar** to B if $B = P^{-1}AP$ for some invertible matrix P.

88 Suppose that A and B be $n \times n$ matrices such that A is similar to B. Prove that $rank\ A = rank\ B$.

$$\text{rank } B = \text{rank } P^{-1}AP$$

$$= \text{rank } P^{-1}A$$

$$= \text{rank } A$$